**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Paper 2: Validation, Testing and Debugging

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does selection do?

2. What is NOT true?

A) true
B) 0.5
C) false
D) An empty string

3. "A Boolean variable represents any unrestricted paragraph of text." Correct this. Explain why.

4. A program joins strings and takes part of an eight-character word. Correct both claims. Explain why. (a) Joining strings adds their numbers. (b) substring(1,3) always means positions 1 to 3 in every language.

## Today's objectives

Explain the core ideas of paper 2: validation, testing and debugging using accurate OCR terminology. Apply a clear method to the worked and guided questions.

## Key vocabulary - Read, cover, recall

**Validation**

Test data against defined rules and loop back after rejection if further input is required.

**Testing**

Include normal, boundary, invalid and erroneous inputs.

**Logic errors**

A syntactically valid program may still use incorrect conditions or updates.

**Maintainability**

Simple named variables, comments and subprograms support future debugging.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Validation** | Test data against defined rules and loop back after rejection if further input is required. |
| **Testing** | Include normal, boundary, invalid and erroneous inputs. |
| **Logic errors** | A syntactically valid program may still use incorrect conditions or updates. |
| **Maintainability** | Simple named variables, comments and subprograms support future debugging. |

## I Do - One worked example

**A test plan is needed for a quantity 1-20 inclusive. Produce four categories.**

1. Normal quantity=10.
2. Boundary quantities=1 and 20.
3. Invalid numeric quantities 0 and 21.
4. Erroneous input 'many' should be handled safely.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Fix a quantity check using quantity>=1 OR quantity<=20.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Fix the range check for an image width of zero pixels.**

```
width = 0
if width >= 1 OR width <= 1920 then
print("Valid")
else
print("Invalid")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

## You Do - Independent exam questions

Q2. Give one invalid and one erroneous input for age 5-16. [2 marks]

Q3. Explain the difference between validation and authentication. [2 marks]

Q4. Describe how a trace table helps locate a logic error. [3 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Boundary testing includes the smallest and largest valid values. ☐ T ☐ F
2. Test data against defined rules and loop back after rejection if further input is required. ☐ T ☐ F
3. Include normal, boundary, invalid and erroneous inputs. ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** It chooses a branch using a condition. Selection can be nested. ☐

**Q2** C - false ☐

**Q3** True or false. ☐

**Q4** (a) Joining 'AB' and '12' gives 'AB12', with no added space. (b) For 'COMPUTER', substring(1,3) in OCR ERL gives 'OMP': indexes start at 0 and 3 is the length. Follow each language's rules. ☐

## Code First - Model

```
width = 0
if width >= 1 AND width <= 1920 then
print("Valid")
else
print("Invalid")
endif
```

**Output: Invalid**

## We Do - Check the model

1. Accept 1 and 20, reject 0 and 21. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Invalid 20 (integer out of range); erroneous 'cat' (non-numeric text). ☐

**Q3** Validation checks input constraints; authentication checks claimed identity. ☐

**Q4** It records successive states and shows the first variable/branch differing from expected behaviour. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**